Tubular & Tubulointerstitial Diseases

Pathology GUE Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty. Attempt every question before checking the answer key on the last page.

4 Easy **10 Medium** **6 Hard**

1. Which of the following is the most common cause of acute kidney injury? EASY

A. Chronic pyelonephritis
B. Acute tubular necrosis
C. Autosomal dominant polycystic kidney disease
D. Renal stones
E. Nephronophthisis

2. Pyuria on urinalysis indicates infection somewhere in the urinary tract, but which additional finding localises the infection to the kidney itself? EASY

A. Hematuria
B. Leukocyte (pus) casts
C. Proteinuria
D. Glucosuria
E. Ketonuria

3. Struvite (ammonium magnesium phosphate) stones form in urine that is: EASY

A. Acidic
B. Alkaline
C. Neutral
D. Highly concentrated but pH-independent
E. Always sterile

4. Which gene is mutated in roughly 85% of autosomal dominant polycystic kidney disease cases? EASY

A. PKD1
B. PKD2
C. PKHD1
D. VHL
E. NPHP1

5. A 45-year-old develops acute kidney injury after a prolonged episode of hypovolemic shock during surgery. Urine microscopy shows necrotic proximal tubular cells with no nuclei, alongside normal distal tubules. This process is best classified as: MEDIUM CASE

A. Chronic obstructive pyelonephritis
B. Acute tubular necrosis due to ischemic injury
C. Acute tubular necrosis due to direct toxic injury
D. Nephronophthisis
E. Medullary sponge kidney

6. A patient on gentamicin develops acute kidney injury with tubular luminal cell attenuation and hyaline/fibrin casts on biopsy. By which mechanism did this drug most likely cause ATN? MEDIUM

A. Ischemic injury via vasoconstriction
B. Direct toxic injury
C. Urinary obstruction
D. Hypersensitivity interstitial nephritis
E. Disseminated intravascular coagulation

7. Which of the following is classified under immunologic causes of tubulointerstitial nephritis in this lecture? MEDIUM

A. Multiple myeloma
B. Heavy metals (lead, cadmium)
C. Sjögren syndrome
D. Chronic urinary tract obstruction
E. Urate nephropathy

8. A woman with recurrent UTIs since childhood is found on imaging to have irregularly scarred kidneys with calyceal blunting, and a history of congenital vesicoureteral reflux. What is the diagnosis? MEDIUM CASE

A. Acute pyelonephritis
B. Chronic reflux-associated pyelonephritis
C. Chronic obstructive pyelonephritis
D. Acute tubular necrosis
E. Medullary sponge kidney

9. A diabetic patient with longstanding urinary obstruction develops acute pyelonephritis complicated by necrosis and sloughing of renal papillae. This complication is: MEDIUM

A. Pyonephrosis
B. Perinephric abscess
C. Papillary necrosis
D. Chronic obstructive pyelonephritis
E. Nephronophthisis

10. A patient with total urinary tract obstruction and acute pyelonephritis develops accumulation of suppurative exudate that is unable to drain, filling the calyces, pelvis, and ureter. This is: MEDIUM CASE

A. Perinephric abscess
B. Papillary necrosis
C. Pyonephrosis
D. Acute tubular necrosis
E. Simple renal cyst

11. A 30-year-old man is found to have hyperuricemia and a radiolucent renal stone. His history includes a myeloproliferative disorder with high cellular turnover. The stone type is most likely: MEDIUM

A. Calcium oxalate
B. Calcium phosphate
C. Ammonium magnesium phosphate
D. Uric acid
E. Cystine

12. A patient with recurrent urease-positive Proteus urinary infections develops a large stone filling the renal pelvis and calyces (a staghorn calculus). What is the composition of this stone? MEDIUM

A. Calcium oxalate
B. Calcium phosphate
C. Ammonium magnesium phosphate (struvite)
D. Uric acid
E. Cystine

13. A neonate is found on ultrasound to have bilaterally enlarged kidneys with a smooth external surface. Biopsy later shows cylindrical and saccular dilations uniformly lined by cuboidal cells. The infant is later found to have periportal fibrosis and bile ductule proliferation on liver biopsy. What is the renal diagnosis? MEDIUM CASE

A. Autosomal dominant polycystic kidney disease
B. Autosomal recessive polycystic kidney disease
C. Multicystic renal dysplasia
D. Medullary sponge kidney
E. Simple renal cyst

14. Compared with PKD1-associated ADPKD, PKD2-associated disease is characterised by: MEDIUM

A. Earlier onset and more rapid progression to renal failure
B. Later onset and slower progression to renal failure
C. No risk of renal failure
D. Exclusively extrarenal manifestations
E. Autosomal recessive inheritance

15. A 50-year-old man on hemodialysis for 10 years is found to have multiple small renal cysts on routine imaging and new-onset hematuria. His glomeruli and tubules show atrophy and scarring on previous biopsy. What is the most important long-term risk associated with his renal findings? HARD CASE

A. Progression to ADPKD
B. Development of congenital hepatic fibrosis
C. Up to a 100-fold increased risk of renal cell carcinoma
D. Struvite staghorn stone formation
E. Conversion to nephronophthisis

16. A renal biopsy from an infant with a multicystic, irregular kidney shows cysts lined by flattened epithelium, with scattered islands of undifferentiated mesenchyme and cartilage. There is no family history of renal disease and the process is unilateral. What is the diagnosis? HARD CASE

A. Autosomal dominant polycystic kidney disease
B. Autosomal recessive polycystic kidney disease
C. Multicystic renal dysplasia
D. Nephronophthisis
E. Acquired cystic disease

17. A patient has multiple small cysts confined to the medulla at the corticomedullary junction, progressive renal insufficiency, and preserved glomeruli on biopsy. There is a family history in a sibling. Which diagnosis best fits, and which variant is most common? HARD

A. Medullary sponge kidney — sporadic variant
B. Nephronophthisis — familial juvenile variant (most common)
C. Nephronophthisis — renal-retinal dysplasia variant (most common)
D. Acquired cystic disease — dialysis-associated
E. Simple renal cyst — solitary variant

18. An adult is incidentally found on CT to have multiple small cysts confined to the renal medulla, with normal renal function and no history of renal failure. The cortex is not involved. What is the most likely diagnosis? HARD CASE

A. Nephronophthisis
B. Medullary sponge kidney
C. Autosomal recessive polycystic kidney disease
D. Acquired (dialysis-associated) cystic disease
E. Multicystic renal dysplasia

19. Which single finding would most reliably distinguish a simple renal cyst from a cystic renal tumour on ultrasonography, per the lecture? HARD

A. Presence of hematuria
B. Smooth contours, avascularity, and a fluid (not solid) signal
C. Bilateral involvement
D. Size greater than 5 cm
E. Presence of a thin capsule

20. A 65-year-old with longstanding hypertension and a 40-pack-year history of chronic pyelonephritis from untreated obstructive uropathy is noted to have corticomedullary scars overlying deformed, blunted calyces, with tubular thyroidisation and periglomerular fibrosis. Which of the following correctly identifies BOTH the underlying mechanism and the expected gross finding? HARD CASE

A. Reflux-associated; smooth, symmetrically contracted kidneys
B. Obstructive; coarse, asymmetric corticomedullary scarring over deformed calyces
C. Acute suppurative; grayish-white cortical abscesses
D. Ischemic ATN; attenuated tubular cells with hyaline casts
E. Autosomal dominant inheritance; bilaterally enlarged kidneys with cysts up to 4 cm

A Answer key & rationale

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	**B**	Acute tubular necrosis is the most common cause of acute kidney injury and is reversible.
2	**B**	Leukocyte (pus) casts form only in the tubules, so their presence localises infection to the kidney.
3	**B**	Struvite stones form in alkaline urine generated by urease-positive organisms such as Proteus vulgaris.
4	**A**	PKD1 accounts for about 85% of ADPKD mutations.
5	**B**	Ischemia (e.g. from hypovolemic shock) causes ATN via vasoconstriction and tubular injury, seen as necrotic PCT with no nuclei beside normal DCT.
6	**B**	Gentamicin is a direct tubular toxin (like heavy metals, myoglobin, and radiation), distinct from the ischemic route.
7	**C**	Sjögren syndrome is listed under immunologic reactions, alongside transplant rejection and sarcoidosis.
8	**B**	Congenital vesicoureteral reflux with renal scarring defines chronic reflux-associated pyelonephritis.
9	**C**	Papillary necrosis is linked to diabetics and patients with urinary tract obstruction.
10	**C**	Pyonephrosis is suppurative exudate that cannot drain due to total obstruction, filling the calyces, pelvis, and ureter.
11	**D**	Hyperuricemia secondary to increased cellular turnover (as in myeloproliferative disorders) produces uric acid stones.
12	**C**	Urease-positive organisms like Proteus and alkaline urine produce ammonium magnesium phosphate (struvite) stones, which form staghorn calculi.
13	**B**	A smooth external surface with cysts at right angles to the cortex, uniform cuboidal cyst lining, and congenital hepatic fibrosis together describe ARPKD.
14	**B**	PKD2 disease has later onset and a slower rise in renal failure risk (15% at 60y, 45% at 70y) compared with PKD1.
15	**C**	Acquired (dialysis-associated) cystic disease carries up to a 100-fold increased risk of renal cell carcinoma.
16	**C**	Islands of undifferentiated mesenchyme or cartilage alongside flattened-epithelium cysts are characteristic of multicystic renal dysplasia.
17	**B**	Corticomedullary-junction cysts with preserved glomeruli and progressive renal insufficiency describe nephronophthisis; the familial juvenile variant is the most common of the three.
18	**B**	Medullary sponge kidney involves small cysts confined to the medulla with normal renal function, usually an incidental finding — the cortex is not involved.
19	**B**	The lecture gives smooth contours, avascularity, and a fluid (not solid) ultrasound signal as what separates a simple cyst from a renal tumour.
20	**B**	Obstruction (renal stone) is the other mechanism of chronic pyelonephritis besides reflux, and its gross hallmark is the coarse, asymmetric corticomedullary scar overlying a deformed calyx described in the lecture.